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Cambridge Ordinary Level

MARK SCHEME for the May/June 2015 series

5054 PHYSICS

5054/42 Paper 4 (Alternative to Practical), maximum raw mark 30

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1	**(a)**	**(i)**	line marked from one shoe <u>to same point</u> on other shoe	B1
		(ii)	40 to 90 cm OR 0.4 to 0.9 m max 2 sf unit required	B1
		(iii)	100 / (a)(ii) in m OR 10 000 / (a)(ii) in cm	B1
	(b)	**(i)**	(push along ground and) count 100 / 200 clicks / turns	B1
		(ii)	sensible comment, e.g. length of step / stride may vary / each 'click' exactly 50 / 100 cm / stride length only an estimate	B1

[Total: 5]

2	**(a)**	**(i)**	1.268 seen	C1
			1.27 s c.a.o. unit required	A1
		(ii)	<u>large</u> variation in raw data / data to 2 d.p. time to fall varies	B1
		(iii)	allows time for parachute to inflate / larger times / more repeatable / minimises percentage error in the time / minimises the <u>effect</u> of (human) reaction error	B1
		(iv)	441(.0) cm^2, c.a.o. unit required correct precision	B1
		(v)	largest square from A4 sheet of paper / sheet 21 (cm) wide / if greater area used, it won't be a square	B1
	(b)(i)(ii)		441 and 1.27 in table <u>with no unit</u> ecf (a)(i)	B1
			400, 324, 256, 196, 144 c.a.o. ecf	B1
		(iii)	axes: correct way round, labelled quantity and unit	B1
			scales: more than ½ grid, linear, not awkward no scales of 3, 7 etc.	B1
			points plotted accurately within ½ small square	B1
			best fit straight line drawn	B1
		(iv)	time needed to fall with no parachute	B1

[Total: 13]

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3 **(a)** correct circuit symbols
all components in a series circuit — B1

(b) A and B labelled at ends of fixed resistor — B1

(c) *y*-shift c.a.o. no additions — B1

(d) **(i)** 3.2 V ± 0.2 unit required — B1

(ii) dot moves up — B1

[Total: 5]

4 **(a)** thermometer
stopwatch / (stop-)clock / timer / watch — B1

(b) diagram of test tube
containing water
thermometer with bulb in water — B1

eye drawn level with top of thread in thermometer — B1

(c) temperature / temp / T / θ AND time / t (or vice versa) — B1

°C / deg C / degree C AND second(s) / s / minutes / min — B1

(d) any two sensible points, e.g.
timer close to test tube / see both together
test tube in clamp stand
thermometer in clamp stand
thermometer with scale facing you
two people with explanation (e.g. count down) — B2
clamp not obscuring the reading
thermometer not touching the sides / bottom of test tube /
⅓ or ½ of thermometer immersed
parallax avoided qualified

[Total: 7]

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